



Presentation Topic: Virtual Network Computing

Course Title: Computer Networks

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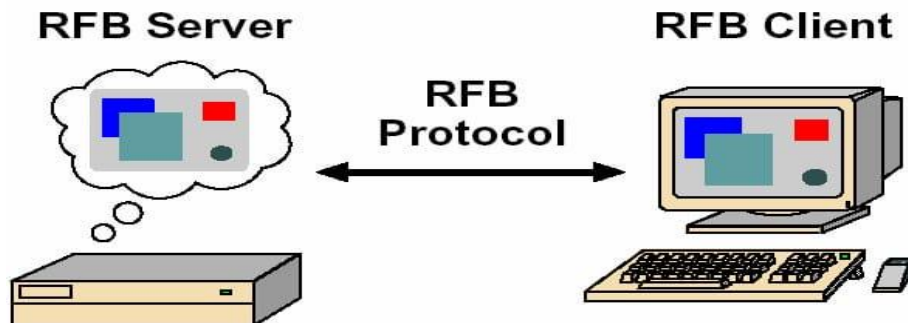
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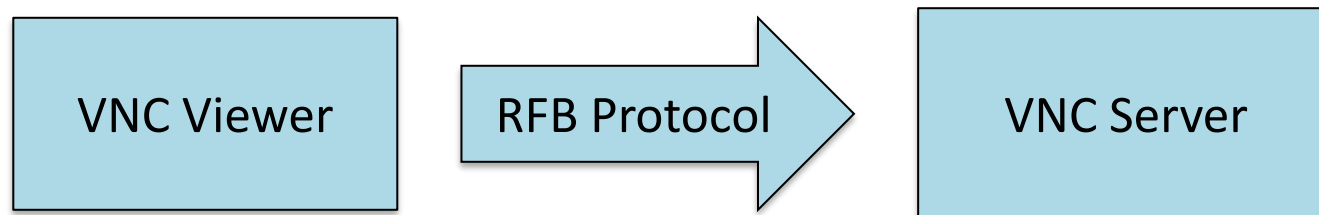
Introduction to VNC

- Virtual Network Computing (VNC) allows remote desktop control.
- It uses the Remote Frame Buffer (RFB) protocol.
- A VNC client connects to a VNC server over a network.
- Commonly used for technical support and remote access.
- Works across Windows, Linux, and macOS platforms.



How VNC Works

- VNC server captures screen images and sends them to the client.
- Client sends back mouse and keyboard inputs.
- Both communicate over TCP/IP protocol.
- Real-time control allows interactive use.
- Used widely in IT management and troubleshooting.



Main Components

- VNC Server: Runs on the computer being controlled.
- VNC Viewer: Installed on the device controlling the server.
- RFB Protocol: Defines communication format.
- Network Connection: Provides link between server and viewer.
- Authentication System: Ensures secure access.



Features of VNC

- Cross-platform compatibility.
- Free and open-source versions available.
- Allows multiple connections to one system.
- Supports clipboard sharing.
- Works with low bandwidth efficiently.

Advantages of VNC

- Easy remote access to systems anywhere.
- Simple setup with minimal configuration.
- No additional hardware required.
- Improves collaboration between users.
- Free and open-source software.

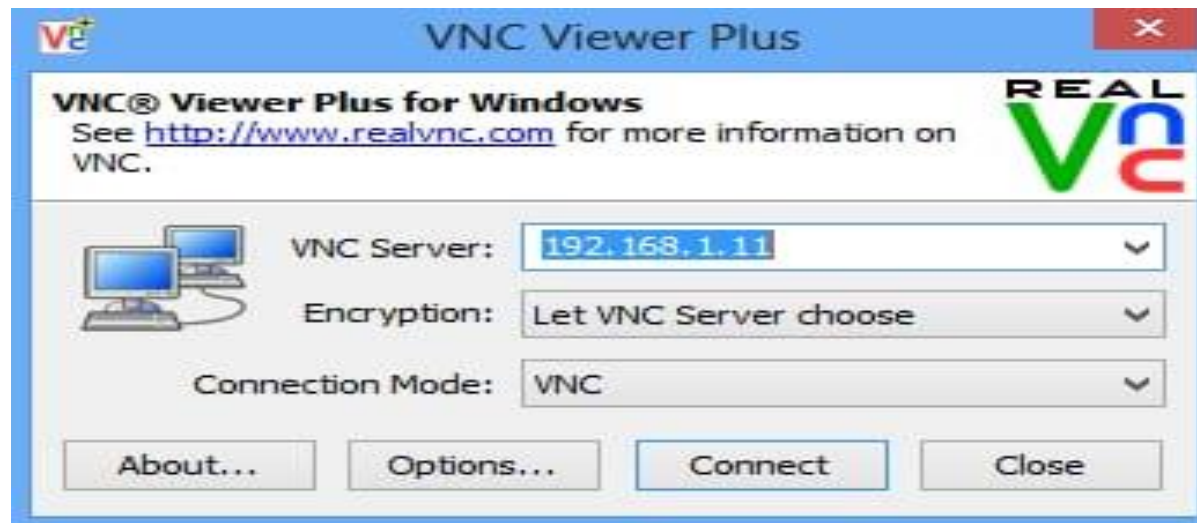


Disadvantages of VNC

- Slower performance on weak connections.
- Limited audio support in basic versions.
- Security depends on configuration.
- May consume more bandwidth during screen refresh.
- Not ideal for heavy multimedia use.

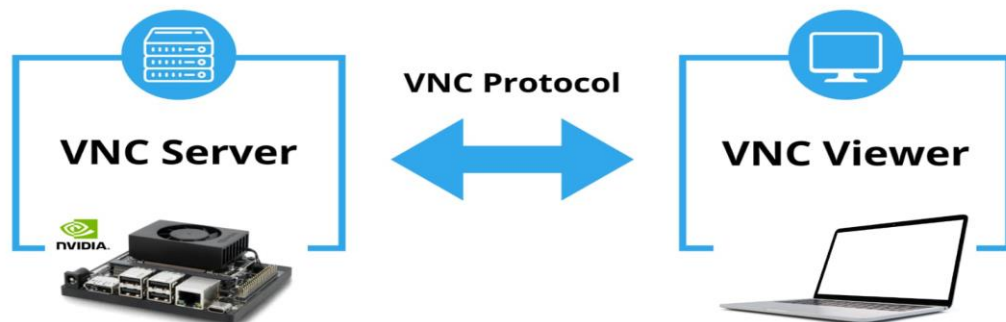
Applications of VNC

- Remote IT support and troubleshooting.
- Accessing office computers from home.
- Online education and demonstrations.
- System administration for data centers.
- Remote collaboration on projects.



VNC Protocol (RFB)

- RFB = Remote Frame Buffer.
- Transmits raw pixel data between devices.
- Uses TCP port 5900 by default.
- Supports encryption and compression.
- Platform-independent and extensible.



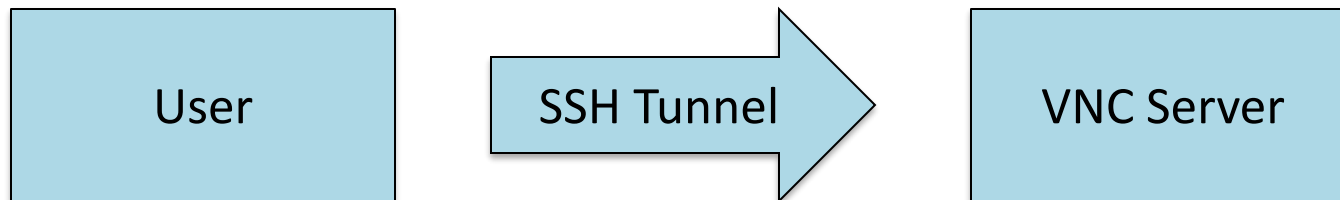
Popular VNC Tools

- RealVNC – Reliable and secure commercial option.
- TightVNC – Lightweight and free.
- UltraVNC – Windows-focused tool with file transfer.
- TigerVNC – High performance and open-source.
- Chrome Remote Desktop – Easy for beginners.



Security in VNC

- Use strong passwords for VNC access.
- Enable encryption when possible.
- Limit access by IP or firewall rules.
- Run through SSH for secure tunnels.
- Always keep VNC software updated.



Future of VNC

- Improved compression for faster performance.
- Enhanced security features with modern encryption.
- Mobile device support and cloud integration.
- Integration with AI for automation.
- Better collaboration tools built-in.

Use Cases in Real Life

- System administrators managing servers remotely.
- Teachers sharing screens during online classes.
- Developers testing software on remote machines.
- Businesses using VNC for client demonstrations.
- Students accessing university lab systems.

Conclusion

- VNC is a powerful tool for remote computing.
- Enables easy control, monitoring, and support.
- Secure, flexible, and cross-platform.
- An essential part of modern IT systems.
- VNC continues to evolve for the future.

